

Solution to Chromatic Contention

Aditya Pahuja

What does such a line look like for $n = 3$ and $n = 4$?

Let a_k and b_k denote chromatopods and chromataceans, respectively, such that a chromatopod a_i and chromatacean a_j are of the same color if and only if $i = j$.

When $n = 3$, a valid line is

$$a_1 a_2 b_1 b_2 a_3 a_2 a_1 b_2 b_1 b_3$$

To show this works, consider what happens when all chromatopods with color 3 are removed (that is, remove a_3 and b_3). The resulting line is

$$a_1 a_2 b_1 b_2 a_2 a_1 b_2 b_1,$$

but then b_2 and a_2 refuse to stand next to each other, so they leave the line, resulting in

$$a_1 a_2 b_1 a_1 b_2 b_1,$$

and so on, eventually reducing the line to nothing; a similar process can be applied to colors 1 and 2 to show that they work as well.

An example of a line that obeys the rules of the ritual for $n = 4$ is

$$a_1 a_2 b_1 b_2 a_3 a_2 a_1 b_2 b_1 b_3 a_4 a_1 a_2 b_1 b_2 a_3 a_2 a_1 b_2 b_1 b_3 b_4.$$

What about arbitrary values of n ?

Let S_i be a sequence satisfying the rules for $n = i$. Then, the sequence

$$S_i a_{i+1} S'_i b_{i+1}$$

also abides by the ritual's constraints, where S'_i is the sequence obtained by reversing S_i , and replacing all the a_k with b_k and vice versa. For example, a possible line S_2 is $a_1 a_2 b_1 b_2$, in which case S'_2 would be

$$a_2 a_1 b_2 b_1.$$

This is very reminiscent of commutators, as a_k and b_k are effectively inverses in this scenario. To see why

$$S_i a_{i+1} S'_i b_{i+1}$$

is a valid line, consider what happens if all creatures with a color corresponding to a value less than or equal to i . S_i entirely disappears by definition, and so does S'_i since chromatopods and chromataceans are interchangeable. This leaves behind $a_{i+1} b_{i+1}$ which goes to nothing, as desired. Alternatively, if the creatures of color $i + 1$ are removed, then S_i and S'_i are inverse sequences, so

they cancel each other out.

It is worth noting that these are not the shortest forms of valid lines, but valid lines in general do take the form of commutators. For example, a shortened variation of the $n = 4$ case is

$$a_1 a_2 b_1 b_2 a_3 a_4 b_3 b_4 a_2 a_1 b_2 b_1 a_4 a_3 b_4 b_3,$$

which is actually a commutator of two commutators!